

Total No. of Questions : 8]

SEAT No. :

**P6638**

[Total No. of Pages : 3

**[6181]-202**

**B.E. (Information Technology)**

**DISTRIBUTED SYSTEMS**

**(2019 Pattern) (Semester - VIII) (414450)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

**Q1) a)** Why is computer clock synchronization necessary? Describe the design requirements for a system to synchronize the clocks in a distributed system. [8]

b) What is NTP? With the help of a diagram, describe how NTP works. [9]

OR

**Q2) a)** Why is the Berkeley algorithm used? Describe how it works using pseudocode. [8]

b) What is MPI? Describe the point-to-point communication in MPI with suitable diagram. [9]

Explain in brief, the following send operations in MPI :

- i) *MPI\_Ssend*
- ii) *MPI\_Bsend,*
- iii) *MPI\_Rsend*
- iv) *MPI\_Isend*

**P.T.O.**

**Q3) a)** What are the key issues in Replica Management? Explain the following with respect to content replication and placement with suitable diagram.[9]

- i) Permanent Replicas
- ii) Server-Initiated Replicas
- iii) Client-Initiated Replicas

b) What is a primary-based protocol in a consistency protocol? Explain the working of primary-backup protocol with suitable diagram. [9]

OR

**Q4) a)** What are the requirements of dependable systems with respect to fault tolerance? How RPC handles the communication failure in the presence of? [9]

- i) The client is unable to locate the server.
- ii) The request message from the client to the server is lost.

b) What is the distribution commit problem? Discuss how this problem is solved using the two-phase commit protocol with suitable diagram. [9]

**Q5) a)** What are the key design issues for distributed file systems? Describe the requirements for distributed file systems. [8]

b) Why Quality of Service Management is important in Distributed Multimedia Systems? Describe QoS manager responsibilities using suitable graphical representation. [9]

OR

**Q6) a)** Describe, using the appropriate diagram, how a web service is implemented in horizontal distribution using web server clusters. [8]

b) Explain in brief, the two places of client-side web caching? Explain cooperative caching with suitable diagram. [9]

**Q7) a)** What is Service Oriented Architecture (SOA)? Explain the various SOA components. How does it differ from traditional software architecture? [9]

b) Explain in brief, the key features of Zabbix. [9]

OR

**Q8) a)** Explain in brief, the key features of Prometheus including data model, query language, or alerting rules. [9]

b) Provide an overview of Mach and CHORUS microkernels. How are memory management techniques used to avoid physical copying of data in Mach and CHORUS? [9]

